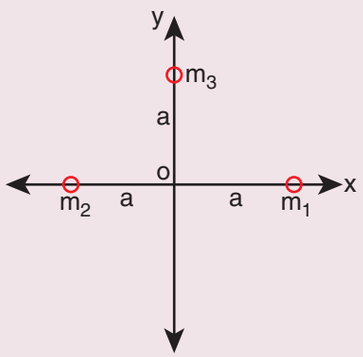
as $G = 6.67 \times 10^{11}$ instead of the correct value $G = 6.67 \times 10^{-11}$, what is the acceleration due to gravity g' for this incorrect G? According to this new acceleration due to gravity, what will be your weight W'?

Ans: g' $= 10^{22}$ g, W' $= 10^{22}$W

10. Calculate the gravitational field at point O due to three masses m_1,m_2 and m_3 whose positions are given by the following figure. If the masses m_1 and m_2 are equal what is the change in gravitational field at the point O?



Ans: $\vec{E} = \frac{G}{a^2}\left[(m_1 - m_2)\hat{i} + m_3\hat{j}\right]$

if $m_1 = m_2, \vec{E} = \frac{G}{a^2}\left[m_3\hat{j}\right]$

11. What is the gravitational potential energy of the Earth and Sun? The Earth to Sun distance is around 150 million km. The mass of the Earth is 5.9×10^{24} kg and mass of the Sun is 1.9×10^{30} kg.

Ans: U= -49.84×10^{32} Joule

12. Earth revolves around the Sun at 30 km s^{-1}. Calculate the kinetic energy of the Earth. In the previous example you calculated the potential energy of the Earth. What is the total energy of the Earth in that case? Is the total energy positive? Give reasons.

Ans: K.E = 26.5×10^{32} J

E = -23.29×10^{32} J

(–) ve implies that Earth is bounded with Sun

13. An object is thrown from Earth in such a way that it reaches a point at infinity with non-zero kinetic energy $\left[\text{K.E}(r = \infty) = \frac{1}{2}\text{M}v_\infty^2\right]$, with what velocity should the object be thrown from Earth?

Ans: $v_e = \sqrt{v_\infty^2 + 2gR_E}$

14. Suppose we go 200 km above and below the surface of the Earth, what are the g values at these two points? In which case, is the value of g small?

Ans: $g_{down} = 0.96$ g

$g_{up} = 0.94$ g

15. Calculate the change in g value in your district of Tamil nadu. (Hint: Get the latitude of your district of Tamil nadu from the Google). What is the difference in g values at Chennai and Kanyakumari?

Ans: $g_{chennai} = 9.7677$ m s^{-2}

$g_{Kanyakumari} = 9.7667$ m s^{-2}

$\Delta g = 0.001$ m s^{-2}

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